Cpts 317. 10/31

hw hint?

Prob 6. Write a grammar for $\{0^n 1^m : n > 2m\}$.

Example words in the lang:

0001, 0000011.

$S \to 0S \mid A$

$A \to 00A1 \mid 0$

$n \geq 2m+1$

Final Exam Ex:

Need a grammar for $\{0^n 1^m : n \geq 3m+2\}$.

do it?

Prob 5. Grammar for

$$\{0^{3n} 1^{2n} 0^{m} 1^{m} : n \geq 1, m \geq 1\}.$$

$S \longrightarrow AB$

$A \rightarrow 000A11 \mid 00011$

$B \rightarrow 0B1 \mid 01$

Prob 8. Need a grammar for

$$\{0^n 1^m : n \neq 2m\}$$

$$= \{0^n 1^m : n > 2m\} \cup$$

$$\{0^n 1^m : n < 2m$$

& we use Prob 6 & Prob 7.

Show: DPDA is strictly weaker PDA.

Proof. We will show

$L = \{w : w = w^r\}$ on $\{a, b\}$ can not be accepted by any DPDA.

Suppose that L can be accepted by a DPDA M.

Ob1. M will not crash on any input word.

Ob2. We construct a PF-tree, with $N_0 = |Q| \times |P|$.

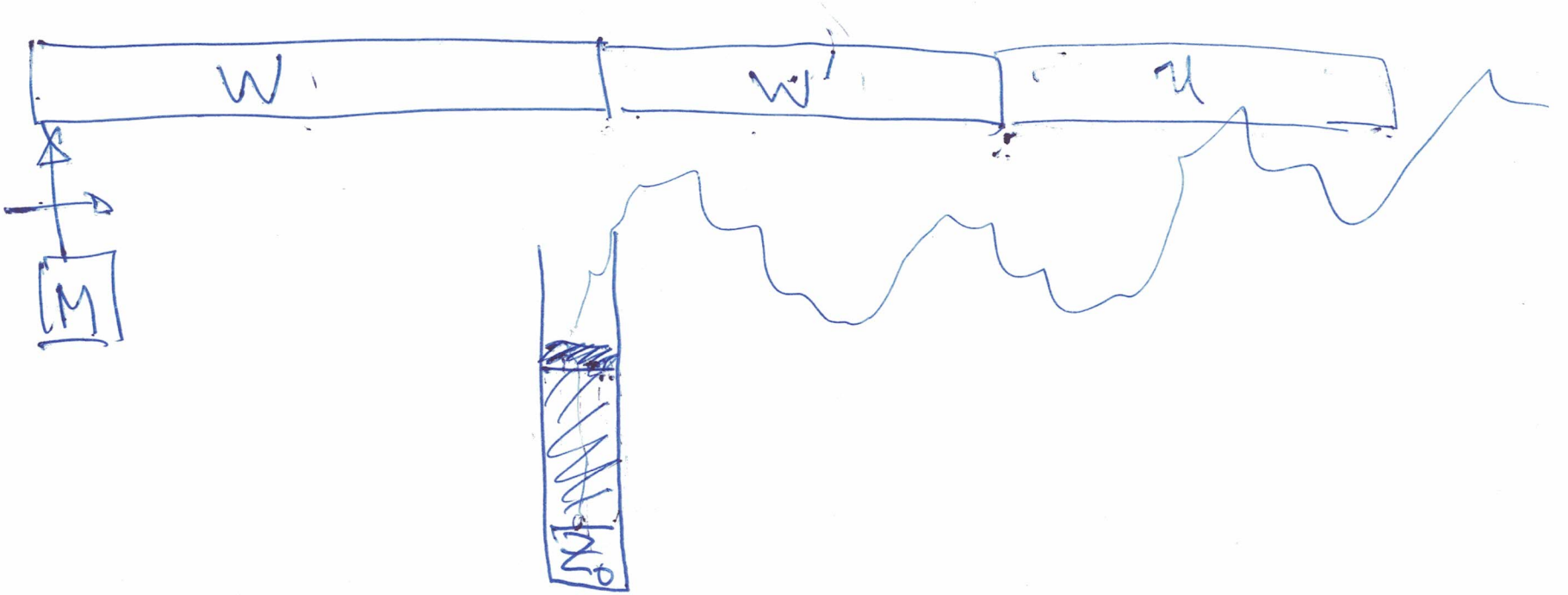
and obtain words:

$$\underbrace{a,\ ba,\ bba,\ bbba,\ \ldots,\ bb\ldots ba,\ b\ldots b}_{N_0+1}.$$

Notice that, in these words, if we pick any two u and v from then, we have

$$u \not\leq v, \quad v \not\leq u.$$

Let w be a word from the PF tree. We can find a word w' such that, when M runs on input ww', at the end of ww'. M will never pop out the top of stack if M continues to run on $ww'u$ for any u.


W
W'
M

Why? Run M on any word ww' with the w as a prefix. At the end of ww', M got to have a stack content. Among all inf. choices of w', M got to have a stack content with lowest stack height for some w'.

Now, if M runs on $ww'u$ for any u then, clearly, the stack top will never be popped out after M reads ww'.

· Recall, we have N_0+1 words for the PF tree:

$$w_1, w_2, \ldots, w_{N_0+1}.$$

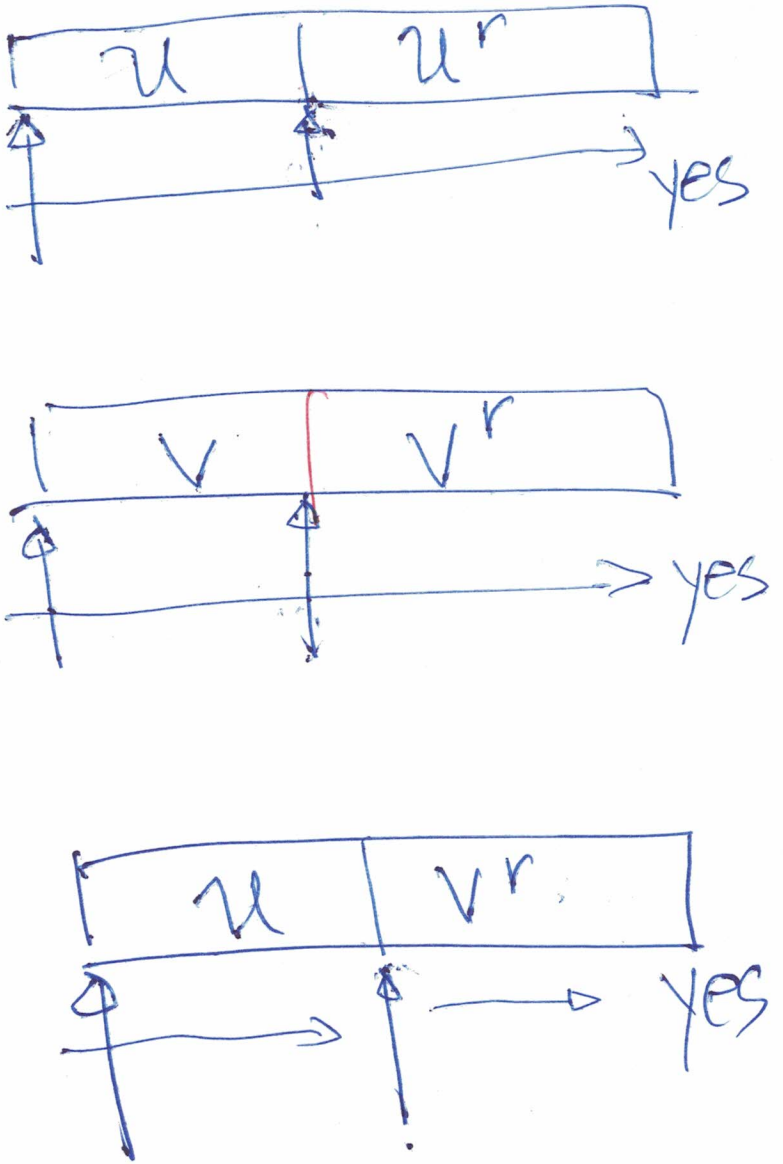
Define $u_1 = w_1 w_1'$.

$$u_2 = w_2 w_2',$$

$$\vdots$$

$$u_{N_0+1} = w_{N_0+1} w_{N_0+1}'.$$

Note that, when M runs on each of the u_i's, M will never pop the stack top out, if it continues to run on any word. Now, for each u_i we use q and γ to denote the state and stack top of M, after M ~~reads all~~



given runs.

reaches the end of u_i. Since $N_0 = |P| \times |Q|$.
we must have two words $u \neq v \in \{u_1, \dots, u_{N_0+1}\}$
such that, the following two runs:

M running on u and

M running on v

will end up with the same state and same stack top. Then, observe that

$uu^r \in L$ and hence M says yes on uu^r.

Similarly, M says yes on vv^r.
We can see that $uv^r \in L$. why? Since M runs on uv^r and M must return yes.

this is contradiction, since

$$uv^r \in L$$

and then $uv^r = vu^r$. this is not possible since $u \not\leq v$, $v \not\leq u$.

$\blacksquare$

$$\text{CFG} \rightleftarrows \text{PDA}$$

$$\text{RegExp} \rightleftarrows \text{FA}$$

Thm. The class of languages accepted by PDA are exactly c.f. languages.

The proof has two parts:

① (Grammar → PDA ; grammar-to-machine construction)

Easy. An algorithm to translate a c.f. grammar into a PDA.

② . (PDA → grammar ; machine-to-grammar construction)

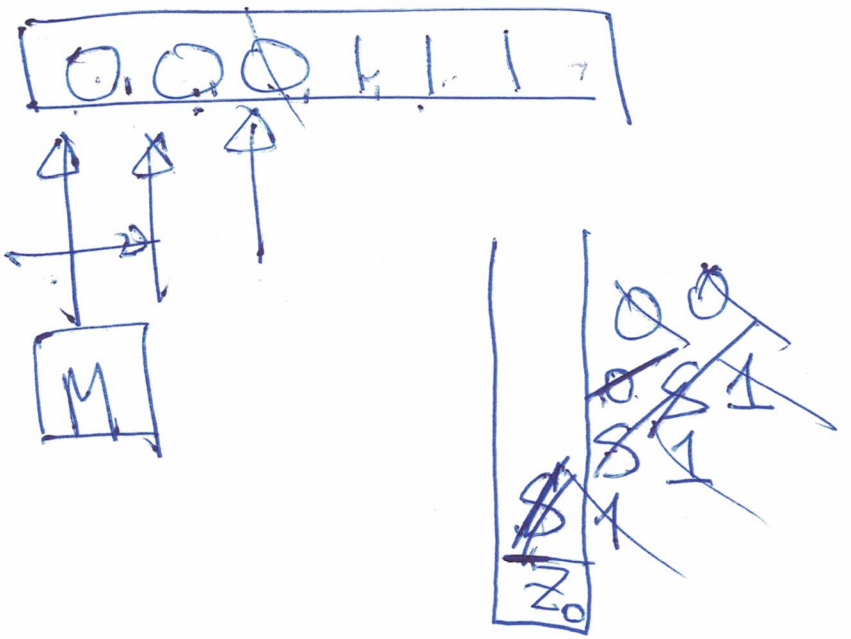
hard An alg to translate a PDA to a c.f. grammar.

Part 1. Grammar → Machine.

Example. grammar G

$$S \longrightarrow 0S1 \mid 01.$$

$S \Rightarrow 0S1$

$\Rightarrow 00S11$

$\Rightarrow 000111$